Code No: 233AB **KR23**

KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

Kmit

(AN AUTONOMOUS INSTITUTE)

Approved by AICTE; Affiliated to JNTUH, Hyderabad-500 029

B.Tech II Year I Semester Supplementary Examinations, July-2025

DATABASE MANAGEMENT SYSTEMS

(COMMON TO CSE, IT, CSE (AI&ML), CSE (DS))

Time: 3 hours Max. Marks: 60

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 10 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b as sub questions.

PART- A (10 Marks)

1. a) Name any two types of database languages. [1M] L1
   b) Define data independence. [1M] L1
   c) What is a primary key? [1M] L1
   d) How do you destroy a table in SQL? [1M] L1
   e) Write the syntax of a basic SELECT query. [1M] L2
   f) What is the purpose of INTERSECT in SQL? [1M] L2
   g) Define normalization? [1M] L1
   h) Expand BCNF? [1M] L1
   i) What is meant by transaction rollback? [1M] L1
   j) What is a schedule in DBMS? [1M] L1

PART- B (5 X 10 =50 Marks)

2. Explain the architecture of a DBMS and describe the different levels of data abstraction. [10M] L2

OR

3. a) Draw an ER-Diagram for Library Management system with minimum four entities. [5M] L2
   b) Justify different types of attributes and relationships which are included in the above ER diagram for Library Management system. [5M] L2

4. Explain the integrity constraints in relational databases with examples. [10M] L2

OR

5. What are views in SQL? Explain the creation, updating, and deletion of views with examples. [10M] L2

6. Write SQL queries for the following operations with syntax and examples. [10M] L2

A. INSERT

B. DELETE

C. UPDATE

D. CREATE.

OR

7. Discuss various types of joins in SQL with examples. [10M] L2

A. INNER JOIN

B. OUTER JOIN

C. LEFT JOIN

D. RIGHT JOIN

E. SELF JOIN

8. Explain 3NF, 2NF and 1NF with suitable examples. [10M] L3

OR

9. Explain the problems caused by redundancy and decomposition in database design. [10M] L2

10. Illustrate Concurrent execution of transaction with examples? [10M] L2

OR

11. What is locking Protocol? Describe the Strict Two Phase locking Protocol. [10M] L2

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Code No: 233AB

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KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

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Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech II Year I Semester Regular Examinations, January-2025

DATABASE MANAGEMENT SYSTEMS

(COMPUTER SCIENCE AND ENGINEERING)

Time: 3 hours Max. Marks: 60

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 10 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b as sub questions.

PART- A (10 Marks)

1. a) What is the need of a data model in DBMS? [1M] L1
   b) Discuss the conceptual design phase in database design. [1M] L1
   c) Explain AUTO INCREMENT constraint with example. [1M] L1
   d) What is the purpose of the GROUP BY clause in SQL queries? Give an example. [1M] L1
   e) Discuss about active databases. [1M] L2
   f) How can you check for NULL values in a WHERE clause? [1M] L1
   g) What is a major consequence of data redundancy in a database? [1M] L1
   h) Why is lossless join decomposition important? [1M] L1
   i) How is atomicity implemented in a database system? [1M] L1
   j) What is the basic idea behind validation-based protocols? [1M] L1

PART- B (5 X 10 =50 Marks)

2. a) Compare and contrast file systems and database management systems. Discuss the advantages and disadvantages of each. [5M] L3
   b) Define the terms "entity," "attribute," and "entity set" in the context of the ER model. Explain the concept of key attributes and their significance. [5M] L2

OR

3. Design and draw an Entity-Relationship (ER) diagram for a Hotel Booking System. [10M] L3
   Your diagram should include:
   - At least three entities with appropriate attributes for each entity.
   - Relationships between these entities.
   - Clearly indicate the cardinalities for each relationship (e.g., 1:1, 1: N, N:M).
   - Ensure that your ER diagram is labeled properly and follows standard conventions.

4. a) What is an integrity Constraint? Discuss the different types of integrity constraints. [5M] L2

b) Consider the following schema: Product (ProductID, Name,Price, Category)

Write an SQL query to ensure:

1. Price is greater than 0.
2. Category is restricted to "Electronics," "Clothing," and "Home Appliances."

Evaluate the role of CHECK constraints in this scenario. [5M] L3

OR

5. Write a SQL query to create a view based on multiple tables and demonstrate querying data from the view. [10M] L3

6. a) Write a SQL query to create a stored procedure that calculates the total marks of a student using IN and OUT parameters. [5M] L3

b) Compare nested query and correlated query. Discuss with suitable examples. [5M] L3

OR

7. Explain about different types of joins with examples. [10M] L2

8. a) Provide an example of a schema that is in 1NF but not in 2NF, and normalize it to 2NF. [5M] L3

b) Explain 5NF with an example relation. [5M] L2

OR

9. a) Discuss the significance of multi-valued dependencies (MVDs) in database design. Provide an example where MVDs cause redundancy and explain how 4NF resolves the issue. [5M] L2

b) Design a schema that avoids redundancy using BCNF for the relation:

Order (OrderID, ProductID, ProductName, SupplierID, SupplierName). [5M] L3

10. a) Demonstrate the use of Save Point, Commit and Rollback. [5M] L3

b) Explain multiple granularity locking in transaction management. [5M] L2

OR

11. a) What is a transaction in a database system? Explain its key properties. [5M] L2

b) Explain the concept of recoverability in transactions and how it ensures database consistency. [5M] L2

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KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

(AN AUTONOMOUS INSTITUTE)

Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech II Year I Semester Regular Examinations, January-2025

DATABASE MANAGEMENT SYSTEMS

(COMMON TO IT, CSE (AI&ML), CSE (DS))

Time: 3 hours **Max. Marks: 60**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 10 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b as sub questions.

PART- A (10 Marks)

1. a) What does the term "schema" represent in a database? [1M] L1
   b) What is the purpose of a weak entity in the ER model? [1M] L1
   c) Discuss the logical database design phase in database design. [1M] L1
   d) Explain the use of HAVING clause in SQL queries with an example. [1M] L2
   e) What is the purpose of the INTERSECT operator in SQL? [1M] L1
   f) What is the difference between an INNER JOIN and a LEFT OUTER JOIN? [1M] L1
   g) Write the need for schema refinement in relational database design. [1M] L2
   h) What is a multi-valued dependency? [1M] L1
   i) Explain what happens to a transaction when it enters the 'aborted' state. [1M] L1
   j) How does the recovery mechanism handle concurrent transactions? [1M] L2

PART- B (5 X 10 =50 Marks)

2. a) Describe the roles and responsibilities of a Database Administrator (DBA). [5M] L2
   b) Explain Database Languages in detail. [5M] L2

OR

3. Design and draw an Entity-Relationship (ER) diagram for a Hospital Management System.
   Your diagram should include:
   - At least three entities with appropriate attributes for each entity.
   - Relationships between these entities.
   - Clearly indicate the cardinalities for each relationship (e.g., 1:1, 1: N, N:M).
   - Ensure that your ER diagram is labeled properly and follows standard conventions. [10M] L3

4. a) Describe the purpose of aggregation operators in SQL. Explain the use of aggregation Operators with examples. [5M] L2
   b) What is a view in a relational database? Explain how views can be used to simplify data access for users. [5M] L2

OR

5. Create a logical database design for a Car Rental System, incorporating all key and integrity constraints. Ensure the creation of appropriate parent tables and child tables, along with their relationships. [10M]L3

6. a) What are nested queries in SQL? Provide an example. [5M] L2

   b) What are triggers in SQL? Explain how they can be used to enforce business rules and maintain data integrity. [5M] L2

OR

7. Design a stored procedure that takes an employee ID as input and returns the employee's name, department name, and manager's name.

Consider the following schema:

- Employee (EmployeeID, EName, MGR, SAL. ,Did)
- Department (DepartmentID, DName, Loc)

Ensure appropriate joins are used to retrieve the required information. [10M] L3

8. a) Define redundancy. What are the common problems caused by redundancy? [5M] L2

   b) Explain the 4NF with an example. [5M] L2

OR

9. a) Provide an example of a schema that is in 2NF but not in 3NF, and normalize it to 3NF. [5M] L3

   b) Define the lossless join property in decomposition. Using an example, explain how to check if a decomposition is lossless. [5M] L2

10. a) Explain the ACID properties and their importance in transactions with examples. [5M] L2

    b) Demonstrate the Timestamp based Concurrency Control. How it is used to ensure Serializability. [5M] L2

OR

11. a) Describe the different states of a transaction during its execution. [5M] L2

    b) Explain and implement a log-based recovery strategy with an example where multiple transactions are involved. [5M] L3

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Code No:224AC **RKR21**

KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

(AN AUTONOMOUS INSTITUTE)

Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech II Year II Semester Supplementary Examinations, June/July -2025

DATABASE MANAGEMENT SYSTEMS

(COMMON TO CSE, IT, CSE(AI&ML), CSE(DS))

Time: 3 hours Max. Marks: 70

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 20 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b as sub questions.

PART- A (20 Marks)

1. a) Identify the Advantages of DBMS. [2M] L2
   b) Elaborate Relationships and Relationship sets. [2M] L2
   c) Show the Differences between Super Key and Candidate Key. [2M] L2
   d) Classify Views and command to create the views. [2M] L2
   e) Summarize Loss less Join Decomposition. [2M] L2
   f) Interpret the Issues raised because of Redundancy. [2M] L2
   g) Identify the need of need of Rollback. [2M] L1
   h) Illustrate Save Point Commit. [2M] L1
   i) Outline creating of users by DBA. [2M] L1
   j) What are the the advantages of triggers. [2M] L1

PART- B (5 X 10 =50 Marks)

2. a) What is data model? Explain Relational Model and E-R model. [5M] L2
   b) Draw an ER-Diagram for Library Management system. [5M] L2

OR

3. a) Demonstrate Conceptual design with the ER Model. [5M] L2
   b) Construct the Structure of DBMS. [5M] L5

4. Define ER model and explain the following kinds of constraints that can be specified in the ER diagram, and give an example of each: i) key constraint ii) participation constraints iii) Domain Constraints [10M] L2

OR

5. a) Explain logical database design with example. [5M] L2
   b) Define View. Explain about fundamental operations. [5M] L1

6. a) What aggregate operators does SQL support ? Explain. [5M] L1

b) Define Functional dependencies and Multi valued dependencies. How are primary keys related to FDs? [5M] L1

OR

7. a) State 1NF, 2NF and 3NF and explain with examples. [5M] L2

b) Interpret Complex Integrity Constraints in SQL. [5M] L4

8. a) Outline the ACID Properties of transactions. [5M] L2

b) What is log file? Explain the following log based recovery schemes.

i) Deferred data base modification ii) Immediate data base modification. [5M] L1

OR

9. a) Classify locking Protocol? Describe the Strict Two Phase Locking Protocol. [5M] L4

b) Show multiple granularity concurrency control scheme. [5M] L1

10. a) Design the methodology for the Execution of stored Procedures from JAVA . [5M] L5

b) Show the importance of Stored Procedures. [5M] L1

OR

11. a) Classify Grant/Revoke permissions on tables using DCL commands. [5M] L4

b) Identify the differences between row-level and statement-level triggers? [5M] L3

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KR21

Code No:214AC

KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

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Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech II Year II Semester Supplementary Examinations, February-2024

Database Management Systems

(COMMON TO CSE, IT, CSE (AI&ML) & CSE (DATASCIENCE))

Time: 3 hours | Max. Marks: 70

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 20 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(2 X 10 =20 Marks)

1. a) List the advantages of DBMS? [2M]L1
   b) Define Strong Entity and give an Example [2M]L1
   c) What is the difference between Delete and Drop [2M]L2
   d) What is the use of CHECK key constraint [2M]L2
   e) Define Aggregate Functions? [2M]L1
   f) What are the problems due to redundancy [2M]L2
   g) Give the states of transaction. [2M]L2
   h) List the phases of two-phase locking protocol [2M]L1
   i) What is stored procedure [2M]L1
   j) What are the responsibilities of DBA [2M]L1

PART- B

(5 X 10 =50 Marks)

2. a) Explain the different levels of Abstractions in DBMS [5M]L2
   b) Summarize the conceptual design with ER model [5M]L2

OR

3. a) What is Data model? List the types of Data models used [3M]L1
   b) Draw an ER-diagram for a banking system? Illustrate different Relationships [7M]L3
   (Hint: Assume at least few entities which cover relationships)

4. Create a student table (roll number, name, branch, age) and branch table (branch name, branch id, block no). When you are creating tables apply key constrains (Primary key, foreign key) and apply DDL queries on the table. [10M]L4

OR

5. a) Illustrate the Importance of key constraints for the creation of table [5M]L2
   b) Compare and Contrast table and View [5M]L4

6. a) Explain DML queries with their syntax and give examples [5M]L2

b) Consider table R (A, B, C, D, E) with FDs as A→B, BC→E and ED→A. [5M]L4

The table is in which normal form? Justify your answer

OR

7. a) What is database schema? Explain the select, project, natural join, union and Cartesian product operations. [6M]L3

b) What is normalization? What is the need for normalization? [4M]L2

8. a) What is a transaction? Explain the ACID properties. [5M]L2

b) Explain Two-Phase Locking protocol. [5M]L2

OR

9. a) Demonstrate conflict serializability with example [5M]L3

b) Explain log-based recovery and mention all its types [5M]L2

10. a) The difference between an "AFTER" trigger and a "BEFORE" Trigger in SQL. [6M]L2

b) Explain different database users. [4M]L2

OR

11. a) Demonstrate how to Create and execute a Stored Procedure with default parameter values. [6M]L3

b) Discuss Grant and Revoke Permissions on a table [4M]L2

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Code No: 214AC

KR21

KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

(AN AUTONOMOUS INSTITUTE)

Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech II Year II Semester Regular Examinations, August - 2023

DATABASE MANAGEMENT SYSTEMS

(COMMON TO IT, CSE (DATASCIENCE) & CSE (AI&ML))

Time: 3 hours | Max. Marks: 70

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 20 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(2 X 10 =20 Marks)

1. a) What is the need of data model in DBMS. [2M]
   b) What is a schema? [2M]
   c) Explain commands with respect to SQL: (i) Rename (ii) Alter [2M]
   d) How to represent a weak entity set in ER diagram. [2M]
   e) What are Assertions? [2M]
   f) Write about transaction operations. [2M]
   g) Define two phase locking mechanism. [2M]
   h) Discuss in brief about outer join. [2M]
   i) Define DBA. [2M]
   j) What are triggers? [2M]

PART- B

(5 X 10 =50 Marks)

2. a) List and explain various data models used for database design. [5M]
   b) Discuss the functionality of query evaluation engine. [5M]

OR

3. What is Entity set and also define Relationship set? List and explain the symbols used to draw ER Diagram with examples. [10M]

4. Consider the following database schema to write queries in SQL Sailor (sid, sname, age, rating) Boats (bid, bname, bcolor), Reserves (sid, bid, day)
   i) Find the sailors who have reserved a red boat
   ii) Find the names of the sailors who have reserved at least two boats
   iii) Find the colors of the boats reserved by 'Mohan'. [10M]

OR

5. Explain the following SQL constructs with examples:
   (1) Order by (2) Group by and having (3) As select (4) Primary key [10M]

P.T.O

6. Explain 3NF & BCNF. What is the difference between them? [10M]

OR

7. a) What is the use of update, drop, and insert operations in DBMS? [5M]
   b) Explain the following Operators in SQL with examples.
      i) UNION ii) INTERSECT iii) EXCEPT [5M]

8. Explain Concurrency control with locking methods. [10M]

OR

9. a) Define transaction and explain desirable properties of transactions. [5M]
   b) What is time stamp ordering? Explain how it is used for concurrency control. [5M]

10. a) Explain about functions and goals of DBA in DBMS. [5M]
    b) Compare the features of in, out and inout parameters with examples? [5M]

OR

11. Discuss in detail the advantages and disadvantages of triggers over stored procedures. [10M]

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Code No: 214AC

KR21

KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

(AN AUTONOMOUS INSTITUTE)

Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech II Year II Semester Regular Examinations, August - 2023

DATABASE MANAGEMENT SYSTEMS

(COMPUTER SCIENCE AND ENGINEERING)

Time: 3 hours | Max. Marks: 70

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 20 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(2 X 10 =20 Marks)

1. a) Define DDL language. [2M]
   b) What are levels of abstraction? [2M]
   c) List any two properties of ER diagram. [2M]
   d) Write a query to create a table with employee details like eno, ename, bdate, address, dno, age, Phone number. [2M]
   e) What is functional dependency? [2M]
   f) Define SECOND normal form. [2M]
   g) What methods are used to assign timestamps to transactions? [2M]
   h) List the ACID properties. [2M]
   i) What is active database? [2M]
   j) List any two differences between a navigational database and a relational database. [2M]

PART- B

(5 X 10 =50 Marks)

2. a) What is DBMS? List four significant differences between file processing system and a DBMS. [5M]
   b) Describe the structure of DBMS with necessary diagram. [5M]

OR

3. Construct an Entity-Relationship diagram for a online shopping systems such as Amazon. [10M]

4. a) What is NULL attribute? With suitable diagram explain weak and strong entity set. [5M]
   b) Explain about various integrity constraints with examples. [5M]

OR

P.T.O

5. Write SQL Queries for following set of tables: EMPLOYEE (EmpNo, Name, DoB, Address, Gender, Salary, DNumber) DEPARTMENT (DNumber, Dname, ManagerEmpNo, ManagerStartDate).

i) Display the Age of 'male' employees.

ii) Display all employees in Department named 'Marketing'.

iii) Display the name of highest salary paid 'female' employee.

iv) Which employee is oldest manger in company? [10M]

6. a) Explain BCNF. Give an example. [5M]

b) Explain about Joins SQL with suitable examples. [5M]

OR

7. What is Normalization? Explain Normal forms with examples. [10M]

8. a) Explain various types of Locks in Detail. [5M]

b) Explain different states of transactions with Diagram. [5M]

OR

9. a) Explain the various TCL commands in detail. [5M]

b) Explain in detail about log based recovery in DBMS. [5M]

10. a) Mention the differences between in, out and inout parameters? [5M]

b) Explain in detail the responsibility of DBA in DBMS. [5M]

OR

11.a) How to implement triggers in SQL explain write example. [5M]

b) Explain about grant and revoke permissions in SQL. [5M]

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Code No:154AB **KR20**

KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

Kmit

(AN AUTONOMOUS INSTITUTE)

Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech II Year II Semester Supplementary Examinations, June/July -2025

DATABASE MANAGEMENT SYSTEMS

(COMMON TO CSE, IT, CSE(AI&ML), CSE(DS))

Time: 3 hours Max. Marks: 75

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b as sub questions.

PART- A (25 Marks)

1. a) List the Database Applications. [2M] L1

   b) What are the types of languages a database system provides? Explain? [3M] L1

   c) Distinguish between Primary key and Foreign key. [2M] L2

   d) What is SQL? Write syntax for Delete command. [3M] L2

   e) Write syntax for Left Outer join. [2M] L1

   f) What is the difference between 3NF and BCNF? [3M] L1

   g) What is transaction? [2M] L1

   h) Explain about durability of transaction. [3M] L2

   i) What is a GRANT and REVOKE command? [2M] L1

   j) Define a View with example. [3M] L1

PART- B (5 X 10 =50 Marks)

2. a) What is data model? Explain Relational Model and E-R model. [5M] L2

   b) Draw an ER-Diagram for Library Management system. [5M] L2

OR

3. a) Identify the main components in a DBMS and briefly explain what they do? [5M] L1

   b) Explain the following: i) View of Data ii) Data Abstraction iii) Instances and Schemas [5M] L2

4. Give an overview of database languages – DDL and DML. [10M] L2

OR

5. Discuss the importance of entity integrity and referential integrity constraints. [10M] L2

6. What is Functional Dependency? Explain types and properties of FD's. [10M] L2

OR

7. What is normalization? Explain 1NF and 2NF Normal forms with examples. [10M] L3

8. Discuss about transaction recovery techniques. [10M] L2

OR

9. How to test serializability of a schedule? Explain with an example. [10M] L2

10. Explain about triggers and discuss the various types with suitable examples. [10M] L2

OR

11. What is stored procedure? Discuss its advantages and provide an example. [10M] L2

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Code No: 154AB **KR20**

KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

(AN AUTONOMOUS INSTITUTE)

Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech II Year II Semester Supplementary Examinations, September-2023

DATABASE MANAGEMENT SYSTEMS

(COMMON TO CSE, IT, CSE (AI&ML)& CSE (DATASCIENCE))

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b, c as sub questions.

PART- A (25 Marks)

1. a) How to represent a Strong and weak entity. [2M]
   b) Write in two to three lines about Data Independence. [3M]
   c) Define about the purpose of Primary Key. [2M]
   d) Name the key that establishes a relationship between two tables, brief about it. [3M]
   e) What is schema refinement [2M]
   f) Define Multi valued dependencies and join dependency [3M]
   g) What is serializability? [2M]
   h) Define timestamp based protocol [3M]
   i) Why a trigger is required in banking sector [2M]
   j) Define the purpose of DBA [3M]

PART- B (5 X 10 =50 Marks)

2. a) Discuss about the various levels of abstraction in DBMS [5M]
   b) Choose any five advantages of DBMS [5M]

OR

3. Explain the importance of ER Diagram while defining a table schema. [10M]

4. a) Give an overview of Data Definition Language [5M]
   b) Discuss in brief about not null, null, foreign Key constraints [5M]

OR

5. a) Discuss in brief about the importance of views [5M]
   b) Explain some tasks that come under DDL? [5M]

6. a) What do you mean by scheme refinement? Explain how it can be accomplished [5M]
   b) Illustrate the problems caused by redundancy and decomposition of relation? [5M]

OR

7. a) Analyze Inner and Outer Joins? [5M]
   b) Explain 4NF, 5NF normal forms with examples. [5M]

P.T.O

8. a) What is transaction? Explain the properties of transaction [5M]

b) Discuss an overview of validation based protocol. [5M]

OR

9. a) Explain about the Multiple granularity Concurrency Control protocol. [5M]

b) Discuss about Lock Base Protocol [5M]

10. a) What is trigger? Explain how to implement triggers in SQL? [5M]

b) Construct Stored Procedures in Java DB with SQL Scripts or JDBC API [5M]

OR

11. a) Give an overview of the DML commands and their importance [5M]

b) Discuss the different types of Database Administers [5M]

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Code No: 154AB **KR20**

KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

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Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech II Year II Semester Supplementary Examinations, APRIL 2023

DATABASE MANAGEMENT SYSTEMS

(COMMON TO CSE, IT, CSE (AI&ML) & CSE (DATASCIENCE))

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b, c as sub questions.

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PART- A (25 Marks)

1. a) Define (i) Database (ii) DBMS. [2M]
   b) Discuss Data Independence. [3M]
   c) Discuss the use of rename operation. [2M]
   d) Explain the purpose of Primary Key. [3M]
   e) Define Aggregate Functions. [2M]
   f) What are the properties of Normalized table? [3M]
   g) Define a checkpoint. [2M]
   h) Explain about transition states. [3M]
   i) Discuss the purpose of trigger. [2M]
   j) What is the purpose of DML commands in SQL? [3M]

PART- B (5 X 10 =50 Marks)

2. a) Compare and Contrast file Systems with database systems. [5M]
   b) Discuss about different types of Data models. [5M]

OR

3. Describe the Structure of DBMS. [10M]

4. Analyze and find whether View exists if the table is dropped from the database. [10M]

OR

5. a) Discuss about Complex integrity constraints in SQL. [5M]
   b) When do you usually use views? [5M]

P.T.O

6. Consider the following relational schema
Employee (empno,name,office,age),Books(isbn,title,authors,publisher) Loan(empno, isbn,date)
Write the following queries using SQL.
   a. Find the names of employees who have borrowed a book Published by McGraw-Hill.
   b. Find the names of employees who have borrowed all books Published by McGraw-Hill.
   c. Find the names of employees who have borrowed more than five different books published by McGraw-Hill.
   d. For each publisher, find the names of employees who have borrowed. **[10M]**

OR

7. Discuss different types of aggregate operators with examples in SQL. **[10M]**

8. Consider the following transactions with data items P and Q initialized to zero:

```
T1: read(P);
    read(Q);
    If P=0 then Q:=Q+1;
    write(Q);
T2: read(Q);
    read(P);
    If Q=0 then P:=P+1;
    write(P);
```

Solve and find any non-serial interleaving of T1 and T2 for concurrent execution leads to a serializable schedule or non-serializable schedule. Explain. **[10M]**

OR

9. Illustrate Concurrent execution of transaction with examples. **[10M]**

10. Explain the Execution process of Stored Procedure from Java. **[10M]**

OR

11. Explain the Responsibilities of DBA. **[10M]**

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KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

(AN AUTONOMOUS INSTITUTE)

Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech II Year II Semester Regular Examinations, August2022

DATABASE MANAGEMENT SYSTEMS

(Common to IT, CSE (AI&ML), CSE (DATA SCIENCE))

Time: 3 hours **Max.Marks: 75**

Answer any five questions

(Each Question Carries 15 Marks)

1. a) What is a data model? Explain in detail about different data models used in Database management systems. [7M]

   b) Classify various features of the ER-Models? How to represent strong entity and week entity set through ER-diagrams. [8M]

2. a) Discuss about Nested and Correlated queries with an example. [7M]

   b) Discuss about various DDL commands with examples. [8M]

3. a) What is decomposition and how does it address redundancy? What problem may be caused by the use of decompositions? [7M]

   b) Explain in detail about 3NF and BCNF. List the differences between 3NF and BCNF [8M]

4. a) By considering an example describe about data update operations in SQL. [7M]

   b) Compute the candidate keys for relation schema R = {A, B, C, D, E} given the functional dependencies F= {A ->BC, CD -> E, B -> D, E -> A }. [8M]

5. a) Explain different types of joins with example queries. [7M]

   b) Discuss different types of aggregate operators with examples in SQL. [8M]

6. a) Explain the importance of Null values in Relational Model. [8M]

   b) Explain logical database design with example. [7M]

7. a) Explain ACID properties of transaction management and Illustrate them through examples. [8M]

   b) Define locking protocol. Explain 2PL and strict 2PL with suitable example. [7M]

8. a) Explain about Stored Procedures with an example. [8M]

   b) What is DBA? Write about DBA Responsibilities. [7M]

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B.Tech II Year II Semester Supplementary Examinations, February-2024

Database Management Systems

(COMMON TO CSE, IT, CSE (AI&ML) & CSE (DATASCIENCE))

Time: 3 hours Max. Marks: 75

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 20 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A (2 X 10 =25 Marks)

1. a) What is Data Independence? [2M]L2
   b) Define Weak Entity and give an Example. [3M]L1
   c) What is varchar2 in DBMS? [2M]L2
   d) Differentiate Table and View related to Data Base. [3M]L2
   e) What is domain integrity? Give example. [2M]L1
   f) List out the Problems related to decompositions. [3M]L2
   g) Define a checkpoint. [2M]L1
   h) List different types of locks. [3M]L1
   i) What are the advantages of stored procedures? [2M]L1
   j) Give DCL Command with their syntax. [3M]L1

PART- B (5 X 10 =50 Marks)

2. a) List the advantages of File system over Data base [5M]L1
   b) Draw an ER-Diagram for a Library system, Assume your own entities (at least 4) and their attributes. [5M]L3

OR

3. a) With a neat sketch explain the structure of DBMS [6M]L2
   b) Distinguish between simple and composite attributes. [4M]L2

4. a) Explain about integrity constraints over relations? [7M]L2
   b) How to create a view and give its importance [3M]L2

OR

5. a) Demonstrate the use of Primary key and Foreign key [5M]L2
   b) Write a short note on logical data base design [5M]L2

6. a) Define Join and explain left outer join with a suitable example. [5M]L2

b) Explain 1NF, 2NF. [5M]L2

OR

7. a) What is nested Query? Give an example. [5M]L2

b) Prove that a relation which is in 4NF must be in BCNF. [5M]L3

8. a) Write a short note on ACID properties. [5M]L2

b) Illustrate concurrency control with lock-based protocols. [5M]L2

OR

9. a) Explain serializability in transaction management. [5M]L2

b) Demonstrate the use of Save Point, Commit and Rollback. [5M]L3

10. a) What is a trigger? Explain with example. [5M]L2

b) Discuss about Database users and Administrators? [5M]L2

OR

11. a) How to create a stored procedure explain with an example. [7M]L3

b) What are the responsibilities of DBA [3M]L2

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B.Tech II Year II Semester Regular Examinations, August2022

DATABASE MANAGEMENT SYSTEMS

(Common to IT, CSE (AI&ML), CSE (DATA SCIENCE))

Time: 3 hours Max.Marks: 75

Answer any five questions

(Each Question Carries 15 Marks)

1. a) What is a data model? Explain in detail about different data models used in Database management systems. [7M]

   b) Classify various features of the ER-Models? How to represent strong entity and week entity set through ER-diagrams. [8M]

2. a) Discuss about Nested and Correlated queries with an example. [7M]

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3. a) What is decomposition and how does it address redundancy? What problem may be caused by the use of decompositions? [7M]

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4. a) By considering an example describe about data update operations in SQL. [7M]

   b) Compute the candidate keys for relation schema R = {A, B, C, D, E} given the functional dependencies F= {A ->BC, CD -> E, B -> D, E -> A }. [8M]

5. a) Explain different types of joins with example queries. [7M]

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6. a) Explain the importance of Null values in Relational Model. [8M]

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   b) Define locking protocol. Explain 2PL and strict 2PL with suitable example. [7M]

8. a) Explain about Stored Procedures with an example. [8M]

   b) What is DBA? Write about DBA Responsibilities. [7M]

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B.Tech II Year II Semester Regular Examinations, August2022

DATABASE MANAGEMENT SYSTEMS

(Computer Science and Engineering)

Time: 3 hours Max.Marks: 75

Answer any five questions

(Each Question Carries 15 Marks)

1. a) Compare the database system with conventional file system. [7M]
   b) List and explain various types of attributes with examples. [8M]
2. a) Explain the importance of avoiding NULL values in a database. [3M]
   b) Write short notes on i) DDL ii) DML iii) Database Schema. [12M]
3. a) Explain about various constraints used in ER-model. [8M]
   b) Differentiate between independent and correlated nested queries. [7M]
4. a)What is normalization? Explain its need. [7M]
   b) Discuss in detail about various normal forms with examples. [8M]
5. a) Explain about Generalization, Specialization and Aggregation with examples. [7M]
   b) What is an integrity Constraint? Discuss about different types of integrity constraints over relations. [8M]
6. Consider the following Relations and answer the queries. [15M]

<--An instance of S3 of Sailors->

sid	sname	rating	age
22	Dustin	7	45.0
29	Brutus	1	33.0
31	Lubber	8	55.5
32	Andy	8	25.5
58	Rusty	10	35.0
64	Horatio	7	35.0
71	Zorba	10	16.0
74	Horatio	9	35.0
85	Arte	3	25.5
95	Bob	3	63.5

<--An Instance of Boats->

Bid	Bname	Color
101	Internlake	Blue
102	Internlake	Red
103	Clipper	Green
104	Marine	Red

(P.T.O)

<- An instance of R2 of Reserves->

sid	bid	day
22	101	10/10/98
22	102	10/10/98
22	103	10/8/98
22	104	10/7/98
31	102	11/10/98
31	103	11/6/98
31	104	11/12/98
64	101	9/5/98
64	102	9/8/98
74	103	9/8/98

a) Find the name and age of the youngest sailor.

b) Find the names of sailors with the highest rating.

c) Find the names of Boats starting with 'C' and ending with 'r'.

d) Write a query to find the total number of boats available.

e) Find the sids of sailors who are having ratings above 7.

7. a) Explain about Conflict Serializability and View Serializability with an example. [8M]

   b) Give an overview of validation-based protocol. [7M]

8. a) Explain about Stored Procedures with an example. [8M]

   b) Explain the basic DCL commands with examples [7M]

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